

EXPT NO:7	IMPLEMENT ADVANCED CHART TYPES USING TABLEAU AND POWER BI
DATE: 14.08.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the importance of comparing actual sales with target sales?

Comparing actual sales with target sales helps an organization understand whether its sales performance is meeting the expected objectives. It shows whether a region is performing above or below its target and helps management identify areas that require improvement. This comparison can also support decisions related to sales strategies, marketing activities, and resource allocation.

2. What is Target Achievement Percentage?

Target Achievement Percentage measures how much of the sales target has been achieved. It is calculated by comparing actual sales with the sales target.

Formula:

$$\text{Target Achievement \%} = (\text{Actual Sales} / \text{Sales Target}) \times 100$$

A value above 100% indicates that the target has been exceeded, while a value below 100% indicates that the target has not been fully achieved.

3. What is Sales vs Target Variance?

Sales vs Target Variance represents the difference between actual sales and the target sales.

Formula:

$$\text{Sales vs Target Variance} = \text{Actual Sales} - \text{Sales Target}$$

A positive variance indicates that actual sales are higher than the target, while a negative variance indicates that actual sales are below the target.

4. Why are maps useful in sales analysis?

Maps are useful because they allow sales performance to be analyzed geographically. They make it easier to identify differences between regions and understand where sales are performing strongly or weakly. In a business dashboard, map visualizations provide a quick geographical view of sales and target-related performance.

5. Why are filters important in an interactive dashboard?

Filters allow users to focus on specific parts of the data without changing the underlying dataset. For example, a region filter can be used to analyze a particular region, while a year filter can be used to examine sales performance for a specific year. This makes the dashboard more interactive and useful for decision-making.

IN-LAB EXERCISE:

OBJECTIVE:

To connect, combine, and model data from multiple sources in Tableau and Power BI and create a unified, comprehensive dashboard for holistic analysis.

PROCEDURE:

1. Data Collection:

- Import two datasets:
 - Dataset 1: Sales Data (CSV) → Order ID, Order Date, Region, Sales Amount
 - Dataset 2: Regional Sales Targets (Excel) → Region, Sales Target, Target Year

2. Data Preparation and Transformation:

- Clean datasets (e.g., standardize "N. America" → "North America").
- Ensure correct data types (numeric for Sales, date for Order Date).
- Handle missing values appropriately.

3. Data Integration and Modeling:

- **Tableau:** Create a join/relationship on the Region field in the Data Source pane.
- **Power BI:** Use the Model view to create a relationship on Region; define cardinality (e.g., many-to-one).

4. Create Calculated Fields for Comparison:

- Target Achievement % = $(\text{SUM}(\text{Sales}) / \text{SUM}(\text{Sales Target})) * 100$
- Sales vs. Target Variance = $\text{SUM}(\text{Sales}) - \text{SUM}(\text{Sales Target})$

5. Design a Comprehensive Dashboard:

- Combine visuals from both sources.
- Add KPI cards: Total Sales, Overall Target Achievement %.
- Use gauge/bullet charts for region-wise target performance.
- Combined bar chart: Actual Sales vs. Target by Region.
- Map visualization: Sales vs. Target Variance by region.

6. Validate Data Integrity:

- Cross-check joined data for accuracy.
- Ensure dashboard filters correctly update visuals.

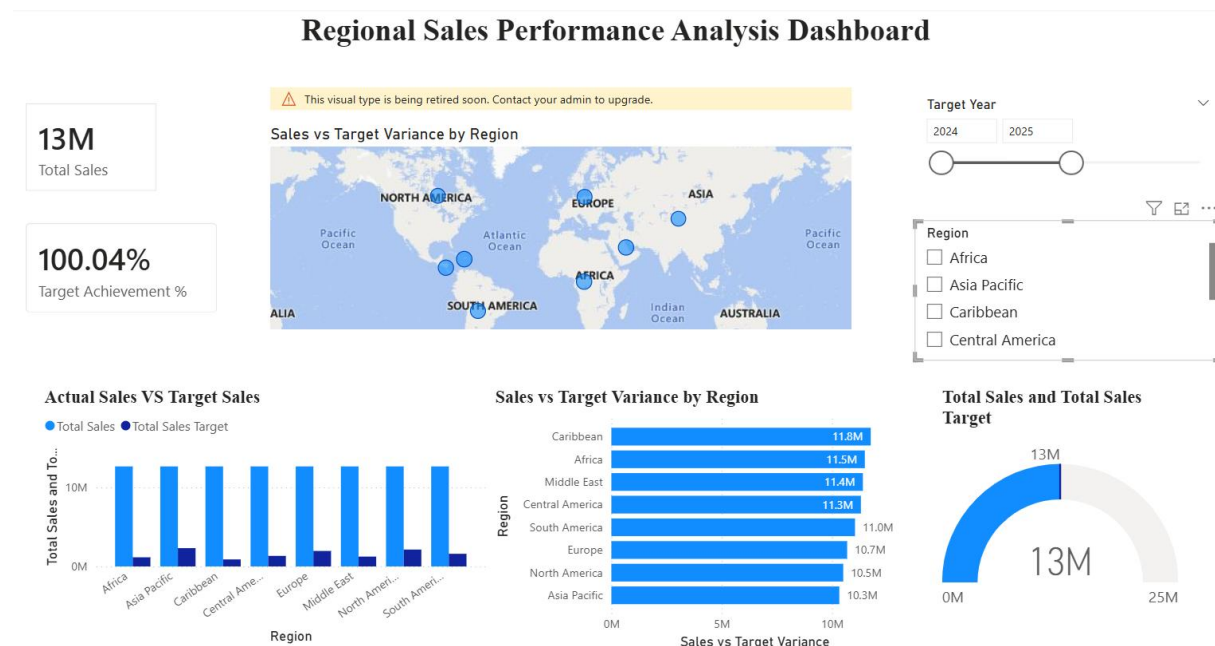
7. Publishing and Sharing:

- **Tableau:** Publish to Tableau Public or Server.
- **Power BI:** Publish to Power BI Service and share.

Experiment Datasets:

- **Dataset 1:** Retail Sales Data (CSV) → Order ID, Order Date, Region, Sales
- **Dataset 2:** Regional Sales Targets (Excel) → Region, Sales Target, Target Year

OUTPUT :



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did integrating the sales targets data provide a more complete analytical picture than using the sales data alone?

Integrating the **Regional Sales Targets** data with the sales data made it possible to compare actual sales performance against the expected targets. Sales data alone shows how much was sold, but it does not indicate whether the sales performance was sufficient to meet the business goal. In the dashboard, the integration allowed us to display Total Sales, Target Achievement %, Actual Sales vs Target Sales, and Sales vs Target Variance by Region. The dashboard shows 13M Total Sales and an overall Target Achievement of 100.04%, indicating that the actual sales slightly exceeded the target.

2. Describe the type of join or relationship you used to combine the two datasets and explain why it was the appropriate choice.

The two datasets were connected using the **Region** field. The Sales Data contained regional sales information, while the Regional Sales Targets dataset contained the corresponding regional sales targets. A relationship based on Region was appropriate because it allowed the sales values and target values to be analyzed together for each region. This relationship enabled the dashboard to compare actual sales with target sales and calculate regional variance and target achievement. The experiment specifically requires creating the relationship on the Region field in Power BI.

3. What challenges, if any, did you face regarding data granularity or mismatched keys (e.g., different region names) between the two sources?

The main consideration during integration was ensuring that the **Region** names were consistent between the two datasets. Since Region was used as the connecting field, inconsistent names could have prevented the sales and target information from matching correctly. After preparing and integrating the data, the dashboard successfully displayed multiple regions such as Africa, Asia Pacific, Caribbean, Central America, Europe, Middle East, North America, and South America. The region filter also displayed these regions, indicating that the regional data was successfully integrated for analysis.

4. How can creating variance calculations (e.g., Sales vs. Target) help a business manager make better decisions?

Variance calculations help managers identify whether actual sales are performing above or below the expected target. In the dashboard, Sales vs Target Variance by Region provides a direct comparison of regional performance. Managers can use this information to identify stronger-performing regions and regions that may require additional attention. The 100.04% Target Achievement shown in the dashboard indicates that overall sales were slightly above the target. Such information can help managers adjust sales strategies, marketing activities, resource allocation, and regional targets. The PDF defines Sales vs Target Variance as the difference between total sales and total sales target.

5. What other data source could you integrate to further enrich this analysis, and what new insights might it provide?

Additional data such as marketing campaign data, customer demographics, product categories, advertising expenditure, inventory data, or competitor pricing could be integrated with the existing sales and target data. For example, marketing campaign data could help determine whether regions with higher promotional activity achieved better sales results. Customer demographic data could provide insights into which customer groups contribute most to regional sales. This would make the dashboard more useful for understanding not only whether a target was achieved, but also the factors influencing regional sales performance. The PDF specifically suggests marketing campaign data and customer demographics as possible additional sources.

RESULT:

The retail sales and regional targets datasets were successfully connected, cleaned, and integrated in Tableau and Power BI. Relationships between the sources created a unified data model. The dashboard visualized sales performance against targets, enabling variance analysis and providing insights into regional performance not possible with a single data source.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		